Advanced Combustion via Microgravity Experiments (ACME)

Radiometer Characterization Report

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PREFACE

Under the Space Flight Systems Development and Operations Contract (SpaceDOC), the National Aeronautics and Space Administration (NASA) has developed a modular, multi-user experimentation facility for conducting fluid physics and combustion science experiments in the microgravity environment of the International Space Station (ISS). This facility, called the Fluids and Combustion Facility (FCF), consists of two test platforms: the Fluids Integrated Rack (FIR) and the Combustion Integrated Rack (CIR). Also included in SpaceDOC are the required support efforts for Mission Integration and Operations, consisting of the Telescience Support Center (TSC) and Mission Integration and Planning (MIP).

The Advanced Combustion via Microgravity Experiments (ACME) is mini-facility designed to support Principal Investigators (PI) interested in studying various aspects of advanced combustion in a microgravity environment. ACME will be integrated into the Combustion Integrated Rack (CIR) onboard the International Space Station in the US Lab Module.

ACME will perform microgravity experiments to study fire suppression/flame extinguishment via 5 experiments: s-flame, Flame design, CLD Flame, E-Field Flames, and Burning Rate Emulator Flames as described in the ACME Baseline Description Document, ACME-DOC-004. ACME will allow investigation using pure as well as mixtures of gaseous fuels at various chamber atmosphere pressures, compositions, and oxygen concentrations. The microgravity environment allows for observation and measurement of flame formation and extinguishment processes that cannot be made in sustained gravity. The modular design of ACME allows for flexibility in configuration for specific experiments and easy maintainability of the hardware.

The core elements of gaseous combustion are included in the ACME design as a basis for current and future gaseous combustion studies. These elements are establishment of gas flow through various exchangeable mass flow controllers, ignition of gas at the burner and the use of appropriate diagnostic measurement devices. The five identified experiments, and any future experiments, use these core elements to build a unique set of requirements for the study of various phenomena distinctive to gaseous combustion. The goal of ACME is to improve the fundamental understanding of combustion processes in order to enable increased efficiency and reduce pollutant emission in practical combustion.

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1.0 INTRODUCTION

1.1 Purpose

The purpose of this report is to document the results for the characterization of the ACME Radiometer Assembly (S1101MFA1520), Flight unit, S/N 1002 and the ACME 60 Degree (Offset) Radiometer (S1101MFA1526), Flight unit, S/N 1002.

1.2 Scope

The scope of this report covers all characterizations, calibrations and tests that were performed in April, 2016 on the ACME flight model. Results and subsequent analysis will be recorded in this final characterization report. This document does not cover all verification test results, but may be used as a reference for verifications.

1.3 Classification

The classification for the ACME Radiometer Assembly and the 60 Degree Radiometer Assembly is Flight Model, including all the hardware listed in Table II.

2.0 DOCUMENTS

This section lists specifications, models, standards, guidelines, handbooks, and other special publications. These documents have been grouped into two categories: applicable documents and reference documents.

2.1 Order of Precedence for Documents

In the event of a conflict between this document and other documents referenced herein, the information of this document shall apply. In the event of a conflict between this document and the contract, the contractual requirements shall take precedence over this document. All documents used, applicable or referenced are to be the issues defined in the Configuration Management (CM) contract baseline. All document changes, issued after baseline establishment, shall be reviewed for impact on scope of work. If a change to an applicable document is determined to be effective and contractually approved for implementation, the revision status will be updated in the CM contract baseline. The contract revision status of all applicable documents is available by accessing the CM database. Nothing in this document supersedes applicable laws and regulations unless a specific exemption has been obtained.

2.2 Applicable Documents

The following documents are applicable to the ACME project to the extent specified herein.

S1101MFA1520 Rev. D	ACME Radiometer Assembly
S1101MFA1526 Rev. A	60 Degree Radiometer Assembly
ACME-REQ-001 Rev. J	ACME Engineering Requirements Document
ACME-TST-022 Rev. A	Radiometer Board Test Procedure
P13001	SpaceDOC Procedure for the Control of Nonconforming Product
P05002	Configuration Management Procedure
P30020	SpaceDOC Safety and Mission Assurance Plan

2.3 Reference Documents

The documents/drawings in this section are provided only as reference material for background information and are not imposed as requirements.

ANSI-ESD S20.20.2007	ESD Association for the Development of an Electrostatic Discharge Program for Protection of Electronic Parts, Assemblies and Equipment (excluding Electrically Initiated Explosive Devices)
ACME-DOC-002 Rev. D	ACME Compliance Matrix
ACME-DOC-007 Rev. A	ACME Concept of Operations
ACME-SRD-001 Rev. -	Integrated Science Requirements Document

3.0 FACILITIES, EQUIPMENT, AND INSTRUMENTATION

3.1 Test Facilities

The ACME flight Radiometer Assembly and 60 Degree Radiometer Assembly Characterization took place in April, 2016 in the ZIN Technologies lab facility located at 6749 Engle Road.

3.2 Equipment and Instrumentation

The following table lists the name and type of the equipment and instrumentation required for the ACME Radiometer and 60 Degree Radiometer Characterization. The calibration status of all instrumentation was verified as appropriate before the characterization began.

TABLE I. Equipment and Instrumentation Information

Equipment	Manufacturer	Model Number	Cal Due Date
DC Regulated Power Supply, 28 VDC	Kenwood	Model: PD36-20AD S/N: 5010016 ZIN Tag: Z000162	N/A (via voltmeter)
Digital Voltmeter/Multi-meter	Fluke	Model: 179 S/N: 81730258 ZIN Tag: Z000371	11 May, 2016
Radiometer Assembly Test Fixture	ZIN Technologies	S1101EGA4583 S/N: 1001	N/A
Laptop w/ Ethernet & Serial Port	Dell	Model: Latitude E5510 S/N: CR8W4N1 ZIN Tag: Z004622	N/A
Oscilloscope	Tektronix	Model: HPLS-36APL750 S/N: 0609520 NASA Tag: 43895	N/A
Blackbody Infrared Source	Infrared Systems Development	Model: IR-564 S/N: 3062K NASA Tag: 09083	7/2016
Blackbody Temp. Controller	Infrared Systems Development	Model: MSO-3014 S/N: C025276 ZIN Tag: 58723	7/2016

4.0 DESCRIPTION OF TEST ARTICLE & CONFIGURATION

4.1 Test Article

4.1.1 ACME Radiometer and 60 Degree Radiometer Assemblies

The ACME Radiometer Assembly (S1101MFA1520) consists of five (5) independent commercial Thermopile detectors (1 - Dexter Research Center, XP-0156 Wide Angle 2M and 4 - Dexter Research Center, XP-0055 Spot 2M) mounted on a single PC board assembly (ACME Radiometer Board Assembly, S1101EFA1524). This assembly is mounted in a single housing (Radiometer Housing Assembly, S1101MFA1521), along with the associated harness for power and data (Radiometer Harness Assembly, S1101EFA1525).

The ACME 60 Degree Radiometer Assembly (S1101MFA1526) consists of a single commercial Thermopile detector (Dexter Research Center, XP-0869 Wide Angle S60M) mounted on a PC board assembly (ACME 60 Degree Radiometer Board Assembly, S1101EFA1527). This assembly is mounted in a housing (60 Degree Radiometer Housing, S1101MFD1528), and is positioned in a fixed location in the ACME Chamber Insert Assembly (CIA) without the need of an associated harness.

The Radiometer Assembly and 60 Degree Radiometer Assembly flight units were assembled in-house by ZIN Technologies at 6749 Engle Road.

TABLE II. Test Article

Description	Manufacturer	Drawing Number	S/N
ACME Radiometer Assembly	N/A	S1101MFA1520	1002
ACME 60 Degree Radiometer Assembly	N/A	S1101MFA1526	1002

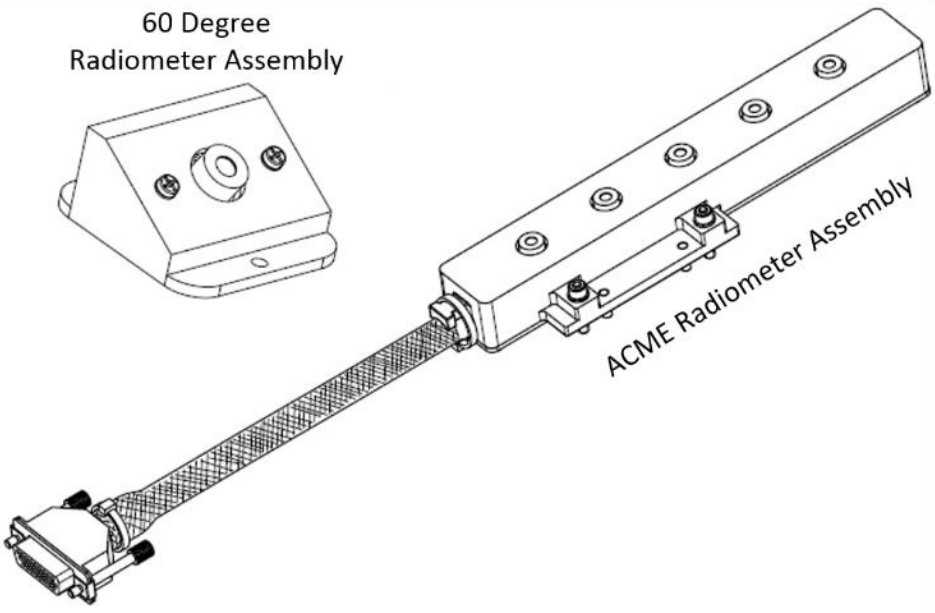


FIGURE 1. ACME Radiometer & 60 Degree Offset Radiometer Assemblies

4.1.1.1 Thermopile Detectors – ACME Radiometer Assembly

Each of the 5 detectors in the ACME Radiometer Assembly is a commercial product made by Dexter Research Center. The Model 2M is a thin film based thermopile detector housed in a TO-5 metal package. The detector element provides a flat spectral response from 100 nm to 100 µm, with an active area of 4.0 mm^2. The back-fill gas is Argon, and the aperture size for the single Wide Angle is 0.202" while the aperture size for the 4 individual Spot Radiometers is 0.080." All other Model 2M performance specifications supplied by Dexter Research Center are listed in the table below.

TABLE III. Dexter Research Center Model 2M Specs

Parameter	Min	Typical	Max	Symbol	Units
Active Area size		2 x 2		AA	mm
Element Area		4		A	mm^2
Number of Junctions		48			
Number of Channels		1			
Output Voltage	200	250	300	V_s	µV
Signal-to-Noise Ratio	12,739	19,531	30,303	SNR	√Hz
Responsivity	15.2	18.9	22.7	ℜ	V/W
Resistance	5	10	15	R	kΩ
Temperature Coefficient of ℜ		-.36			%/°C
Temperature Coefficient of R		-.2			%/°C
Noise Voltage	9.9	12.8	15.7	V_n	nV/√Hz
Noise Equivalent Power	.44	.68	1.0	NEP	nW/√Hz
Detectivity	1.9	3.0	4.6	D*	10^8cm√Hz/W
Time Constant		85		τ	ms
Field of View		38°/95°		FOV	Degrees
Package Type		TO-5			
Operating Temperature	-50		100	T_a	°C

4.1.1.2 Thermopile Detector – 60 Degree Radiometer Assembly

The single detector in the 60 Degree Radiometer Assembly is a commercial product made by Dexter Research Center. The Model S60M is a silicon based thermopile detector housed in a TO-5 metal package. The detector element provides a flat spectral response from 100 nm to 100 µm, with an active area of 0.36 mm^2. The back-fill gas is Argon, and the aperture size for this Wide Angle view is 0.092." All other Model S60M performance specifications supplied by Dexter Research Center are listed in the table below.

TABLE IV. Dexter Research Center Model S60M Specs

Parameter	Min	Typical	Max	Symbol	Units
Active Area size		.6 x .6		AA	mm
Element Area		.36		A	mm^2
Number of Junctions		72			
Number of Channels		1			
Output Voltage	90	120	140	V_s	µV
Signal-to-Noise Ratio	2,123	3,125	4,294	SNR	√Hz
Responsivity	75.8	101.0	117.8	ℜ	V/W
Resistance	65	90	120	R	kΩ
Temperature Coefficient of ℜ		-.04			%/°C
Temperature Coefficient of R		.11			%/°C
Noise Voltage	32.6	38.4	42.4	V_n	nV/√Hz
Noise Equivalent Power	.28	.38	.56	NEP	nW/√Hz
Detectivity	1.1	1.6	2.2	D*	10^8cm√Hz/W
Time Constant		32		τ	ms
Field of View		64°/81°		FOV	Degrees
Package Type		TO-5			
Operating Temperature	-50		100	T_a	°C

4.1.1.3 Thermopile Window Material

The Window material selected for the detectors used is Barium Fluoride (BaF_2). This material selection is common to all of the thermopiles in both the ACME Radiometer Assembly and the 60 Degree Radiometer Assembly. Barium Fluoride has a typical average Transmission of 91% over a wavelength range of 0.2 – 17.5 µm. The transmission curve is shown in the figure below.

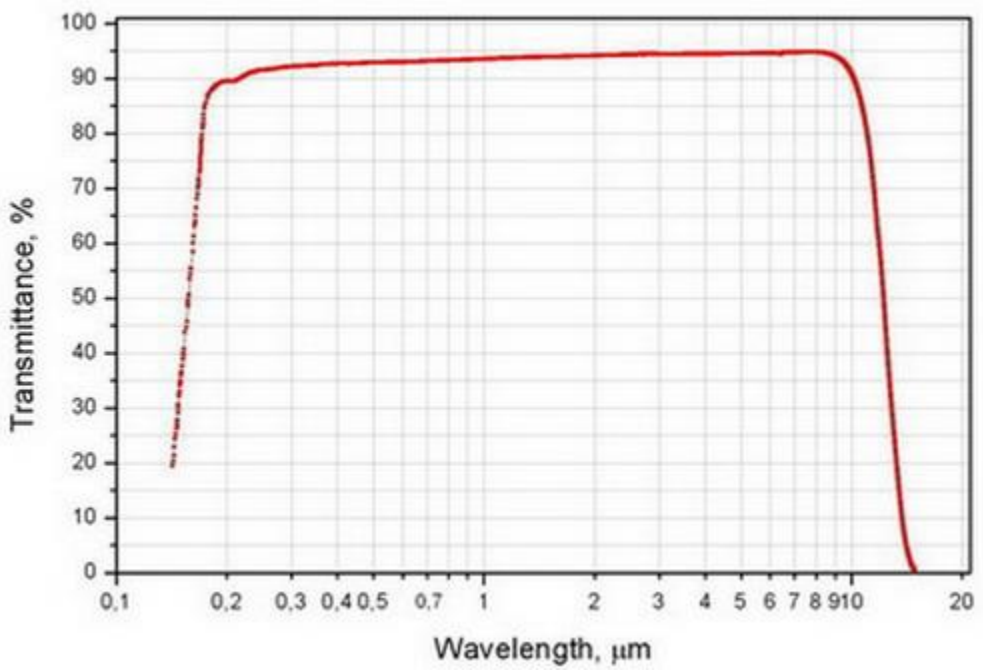


FIGURE 2. Barium Fluoride Transmission Curve

4.1.1.4 Test Configuration

ACME radiometers are used for temperature based measurements, thus a calibrated Blackbody temperature source was used for this characterization.

Each of the Radiometer Assemblies being characterized and the Blackbody source are positioned on an optical bench. The Blackbody source output was aligned, via optical mounting hardware, with the opening of the particular Radiometer detector location being characterized.

The optical mounting hardware allows for alignment of each Radiometer Assembly in both the X and Y directions and also sets the distance from the outer surface of each detector to the Blackbody aperture wheel (Z direction), equal to the appropriate flight distance: 138.5 mm for the ACME Radiometer Assembly and 157.5 mm for the 60 Degree Radiometer Assembly.

All testing was done under blackout curtains that were arranged such that only the output from the Blackbody was visible during characterization. See the picture below for the full setup (Note: side curtains are raised in order to obtain the picture).

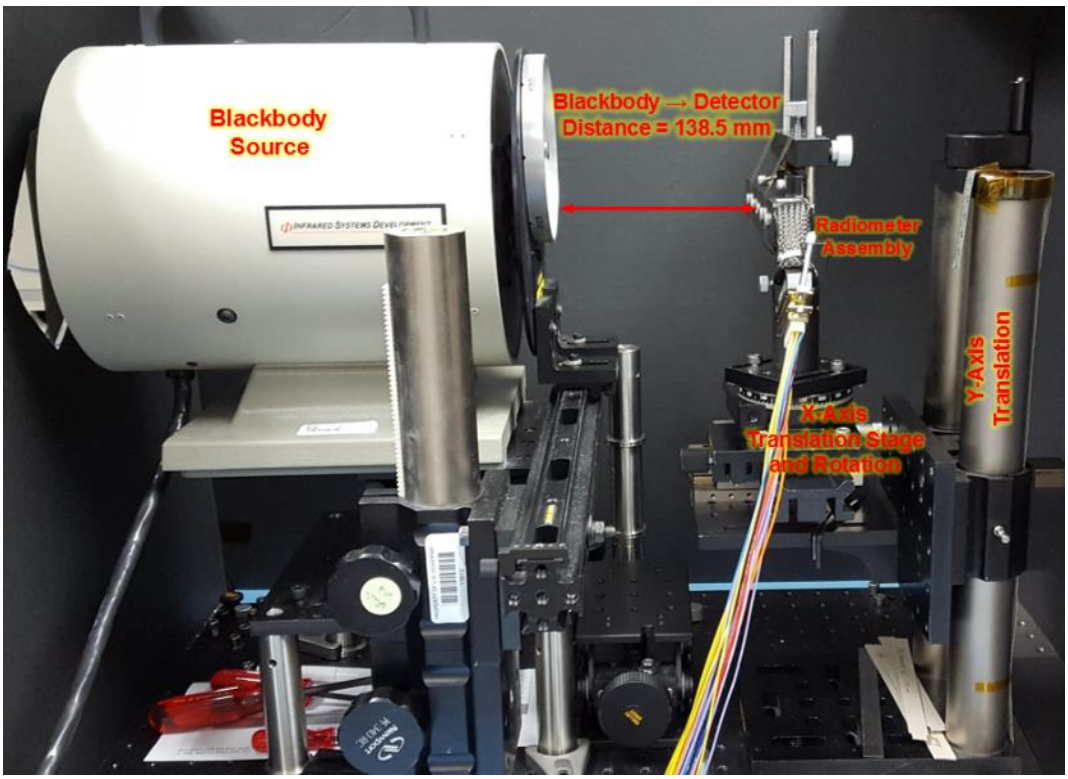


FIGURE 3. Radiometer Assembly Alignment Setup

To reduce the error caused by the Blackbody Aperture Wheel heating up over time and emitting unintended radiation, all Blackbody characterizations in this document were run with the Aperture Wheel mounted it on a slide rail, allowing it to be retracted away from the hot Blackbody source between measurements. By limiting the time resident near the hot Blackbody source, the aperture wheel could be held at room temperature.

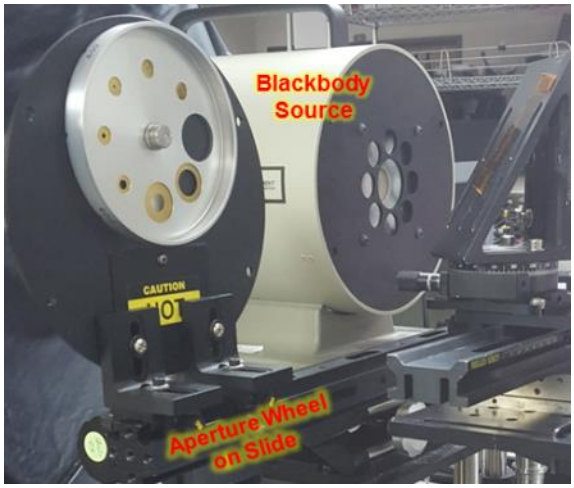


FIGURE 4. Aperture Wheel on Translation Slide

Each Radiometer Assembly is connected to the Radiometer Assembly Test Fixture for both +5 VDC power in and output (V_{out}) readings as shown in the figure below. Further details on the specific connections can be found in ACME-TST-022, "Radiometer Board Test Procedure."

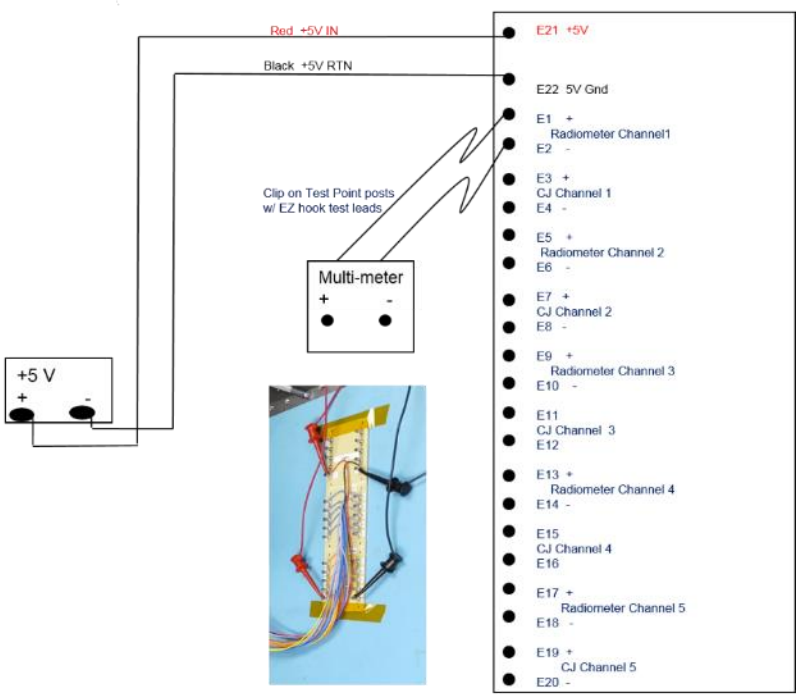


FIGURE 5. Test Fixture for Characterization Setup

A voltmeter was used to monitor the output signal (V_{out}) of each individual detector in the assembly. An additional voltmeter was connected to the Power Supply to ensure the input voltages remained within the required +5 VDC range. The DC Power Supply, Voltmeters, Oscilloscope and Blackbody Temperature Controller are located on a bench adjacent to the Radiometer Assembly and Blackbody (on the optical bench behind this shelf). Below is a photo of the support equipment in the laboratory setting.



FIGURE 6. GSE for Characterization Setup

5.0 ACME RADIOMETER ASSEMBLY CHARACTERIZATION

5.1 ACME Radiometer Assembly Field of View

5.1.1 FOV Setup

The Field of View (FOV) measurement is performed separately for each of the 5 thermopile detectors in the ACME Radiometer Assembly. Each detector is centered and aligned normal to the Blackbody source, at a distance of 138.5 mm from the Blackbody Aperture to the outer face. Each detector is then translated in the horizontal (X) axis over a range of ±70 mm about the center position, in 10 mm increments, while recording the output signal level. This determines the X axis FOV and roll-off characteristics.

While positioned at the X axis center point for that detector, the vertical (Y) axis characterization is measured by raising/lowering the assembly on a lab jack over ±70 mm to measure the Y axis FOV and roll-off characteristics.

5.1.2 Center – Wide Angle Radiometer Field of View Data / Analysis

Prior to the measurements, the Center detector was aligned to the Blackbody source, by locating the peak of the signal output at the current Gain setting. The resulting Center detector response to off-axis translations is shown in the figure below. It shows the X (horizontal, along the axis parallel to the line bisecting the detector apertures) and Y (vertical, along the axis perpendicular to the line bisecting the detector apertures) FOV and roll-off measurements.

The following curves show the results from the FOV scans in both the X and Y directions for Center – Wide Angle Radiometer.

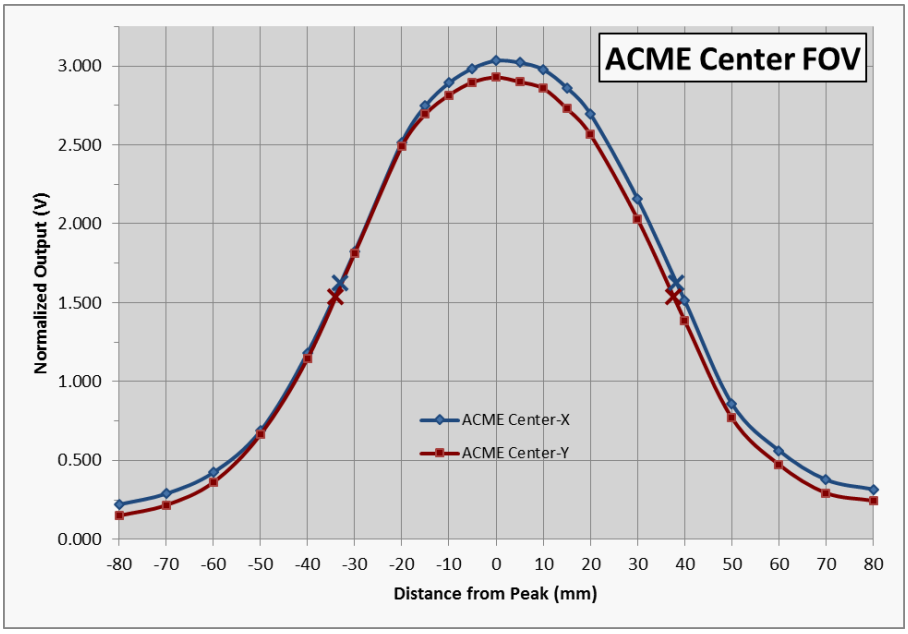


FIGURE 7. Center Wide Angle – FOV Characterization

The following parameter settings were used for the Center Wide Angle FOV scans:

Blackbody Temperature = 750°C
Blackbody Aperture = 0.4”

The following table shows the data resulting from the FOV scans for the Center – Wide Angle Radiometer.

TABLE V. Data for Center – Wide Angle FOV

Center - X Stats:			Center - Y Stats:		
P-P	2.812	V	P-P	2.777	V
V_{FWHM}	1.627	V	V_{FWHM}	1.539	V
X1	-33.1	mm	Y1	-34.1	mm
X2	38.2	mm	Y2	37.6	mm
FWHM	71.3	mm	FWHM	71.8	mm
$Center_{FWHM}$	**2.6**	**mm**	**$Center_{FWHM}$**	**1.8**	**mm**
$V_{90\%}$	2.752	V	$V_{90\%}$	2.649	V
$X1_{90\%}$	-14.8	mm	$Y1_{90\%}$	-16.1	mm
$X2_{90\%}$	18.3	mm	$Y2_{90\%}$	17.4	mm
$FW_{90\%}$	33.2	mm	$FW_{90\%}$	33.5	mm
$Center_{Alignment}$	**1.7**	**mm**	**$Center_{Alignment}$**	**0.7**	**mm**
V_{FOV}	0.502	V	V_{FOV}	0.428	V
$X1_{FOV}$	-57.0	mm	$Y1_{FOV}$	-57.8	mm
$X2_{FOV}$	63.3	mm	$Y2_{FOV}$	62.6	mm
FOV	**120.3**	**mm**	**FOV**	**120.4**	**mm**
$Center_{FOV}$	3.1	mm	$Center_{FOV}$	2.4	mm

The results for Center – Wide Angle Radiometer show that the center of symmetry does not completely agree with the geometric peak position, as indicated by the X_{center} and Y_{center} fits of -2.6 mm and 1.8 mm, respectively. These centers were calculated by determining the Full Width at Half Maximum (FWHM) points on each curve and interpolating to get the position of each FWHM point.

The 90% points (10% roll-off) were used to verify the alignment of the detector to the measurement location:

Center – Wide Angle Alignment: X = 1.7 mm and Y = 0.7 mm

The 10% points were used to determine the overall Field of View:

Center – Wide Angle Radiometer FOV: X = 120.3 mm and Y = 120.4 mm

5.1.3 +30 mm – Spot Radiometer Field of View Data / Analysis

Prior to the measurements, the +30 mm Spot detector was aligned to the Blackbody source, by locating the peak of the signal output at the current Gain setting. The resulting detector response to off-axis translations is shown in the figure below. It shows the X (horizontal, along the axis parallel to the line bisecting the detector apertures) and Y (vertical, along the axis perpendicular to the line bisecting the detector apertures) FOV and roll-off measurements.

The following curves show the results from the FOV scans in both the X and Y directions for the +30 mm Spot Radiometer.

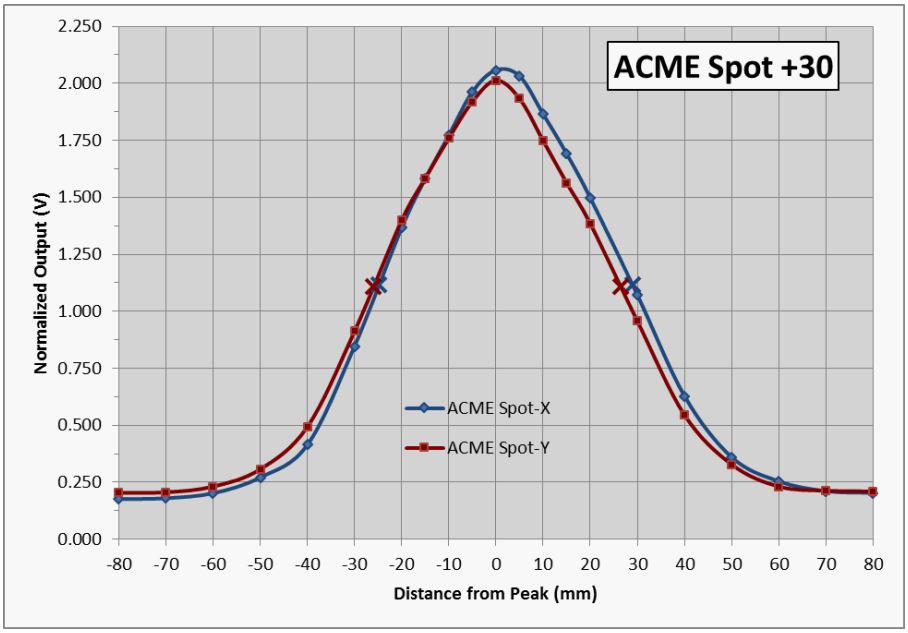


FIGURE 8. +30 mm Spot – FOV Characterization

The following parameter settings were used for the +30 mm Spot FOV scans:

Blackbody Temperature = 850°C
Blackbody Aperture = 0.2”

The following table shows the data resulting from the FOV scans for the +30 mm Spot Radiometer.

TABLE VI. Data for +30 mm Spot FOV

+30 - X Stats:			+30 - Y Stats:		
P-P	1.882	V	P-P	1.809	V
V_{FWHM}	1.117	V	V_{FWHM}	1.109	V
X1	-24.8	mm	Y1	-26.0	mm
X2	28.9	mm	Y2	26.5	mm
FWHM	53.7	mm	FWHM	52.4	mm
$Center_{FWHM}$	**2.1**	**mm**	**$Center_{FWHM}$**	**0.2**	**mm**
$V_{90\%}$	1.870	V	$V_{90\%}$	1.832	V
$X1_{90\%}$	-7.4	mm	$Y1_{90\%}$	-7.7	mm
$X2_{90\%}$	9.9	mm	$Y2_{90\%}$	7.7	mm
$FW_{90\%}$	17.3	mm	$FW_{90\%}$	15.5	mm
$Center_{Alignment}$	**1.2**	**mm**	**$Center_{Alignment}$**	**0.0**	**mm**
V_{FOV}	0.364	V	V_{FOV}	0.385	V
$X1_{FOV}$	-43.5	mm	$Y1_{FOV}$	-45.8	mm
$X2_{FOV}$	49.8	mm	$Y2_{FOV}$	47.4	mm
FOV	**93.3**	**mm**	**FOV**	**93.2**	**mm**
$Center_{FOV}$	3.2	mm	$Center_{FOV}$	0.8	mm

The results for the +30 mm Spot Radiometer show that the center of symmetry does not completely agree with the geometric peak position, as indicated by the X_{center} and Y_{center} fits of 2.1 mm and 0.2 mm, respectively. These centers were calculated by determining the Full Width at Half Maximum (FWHM) points on each curve and interpolating to get the position of each FWHM point.

The 90% points (10% roll-off) were used to verify the alignment of the detector to the measurement location:

+30 mm Spot Alignment: X = 1.2 mm and Y = 0.0 mm

The 10% points were used to determine the overall Field of View:

+30 mm Spot Radiometer FOV: X = 93.3 mm and Y = 93.2 mm

5.1.4 –30 mm Spot Radiometer Field of View Data / Analysis

Prior to the measurements, the -30 mm Spot detector was aligned to the Blackbody source, by locating the peak of the signal output at the current Gain setting. The resulting detector response to off-axis translations is shown in the figure below. It shows the X (horizontal, along the axis parallel to the line bisecting the detector apertures) and Y (vertical, along the axis perpendicular to the line bisecting the detector apertures) FOV and roll-off measurements.

The following curves show the results from the FOV scans in both the X and Y directions for the -30 mm Spot Radiometer.

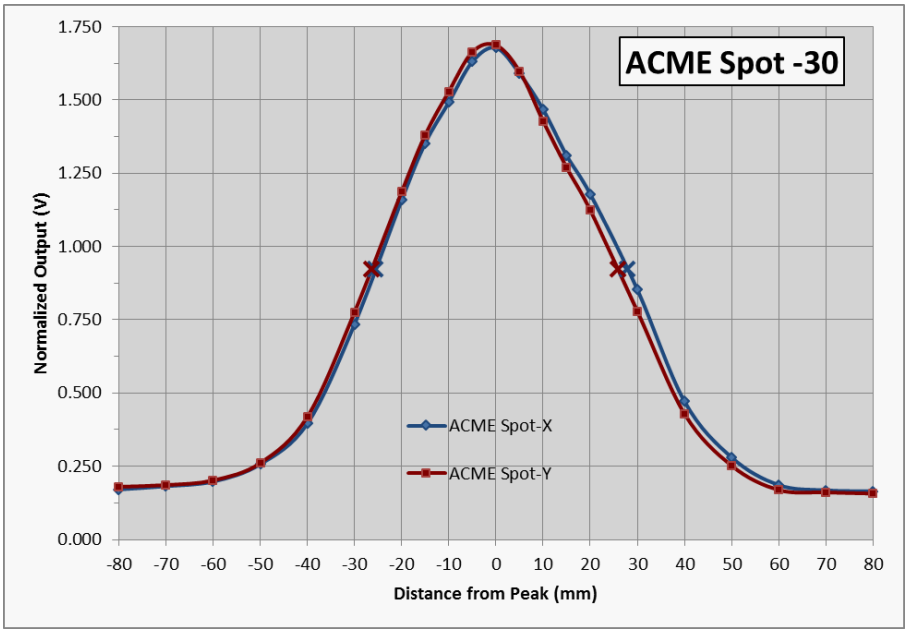


FIGURE 9. –30 mm Spot – FOV Characterization

The following parameter settings were used for the -30 mm Spot FOV scans:

Blackbody Temperature = 900°C
Blackbody Aperture = 0.2”

The following table shows the data resulting from the FOV scans for the -30 mm Spot Radiometer.

TABLE VII. Data for –30 mm Spot FOV

-30 - X Stats:			-30 - Y Stats:		
P-P	1.513	V	P-P	1.530	V
V_{FWHM}	0.922	V	V_{FWHM}	0.922	V
X1	-25.6	mm	Y1	-26.4	mm
X2	27.9	mm	Y2	25.9	mm
FWHM	53.5	mm	FWHM	52.3	mm
$Center_{FWHM}$	**1.2**	**mm**	**$Center_{FWHM}$**	**-0.3**	**mm**
$V_{90\%}$	1.527	V	$V_{90\%}$	1.534	V
$X1_{90\%}$	-8.8	mm	$Y1_{90\%}$	-9.8	mm
$X2_{90\%}$	7.6	mm	$Y2_{90\%}$	6.9	mm
$FW_{90\%}$	16.3	mm	$FW_{90\%}$	16.7	mm
$Center_{Alignment}$	**-0.6**	**mm**	**$Center_{Alignment}$**	**-1.5**	**mm**
V_{FOV}	0.316	V	V_{FOV}	0.310	V
$X1_{FOV}$	-45.7	mm	$Y1_{FOV}$	-46.9	mm
$X2_{FOV}$	48.1	mm	$Y2_{FOV}$	46.7	mm
FOV	**93.8**	**mm**	**FOV**	**93.6**	**mm**
$Center_{FOV}$	1.2	mm	$Center_{FOV}$	-0.1	mm

The results for the -30 mm Spot Radiometer show that the center of symmetry does not completely agree with the geometric peak position, as indicated by the X_{center} and Y_{center} fits of 1.2 mm and -0.3 mm, respectively. These centers were calculated by determining the Full Width at Half Maximum (FWHM) points on each curve and interpolating to get the position of each FWHM point.

The 90% points (10% roll-off) were used to verify the alignment of the detector to the measurement location:

-30 mm Spot Alignment: X = -0.6 mm and Y = -1.5 mm

The 10% points were used to determine the overall Field of View:

- 30 mm Spot Radiometer FOV: X = 93.8 mm and Y = 93.6 mm

5.1.5 +60 mm Spot Radiometer Field of View Data / Analysis

Prior to the measurements, the +60 mm Spot detector was aligned to the Blackbody source, by locating the peak of the signal output at the current Gain setting. The resulting detector response to off-axis translations is shown in the figure below. It shows the X (horizontal, along the axis parallel to the line bisecting the detector apertures) and Y (vertical, along the axis perpendicular to the line bisecting the detector apertures) FOV and roll-off measurements.

The following curves show the results from the FOV scans in both the X and Y directions for the +60 mm Spot Radiometer.

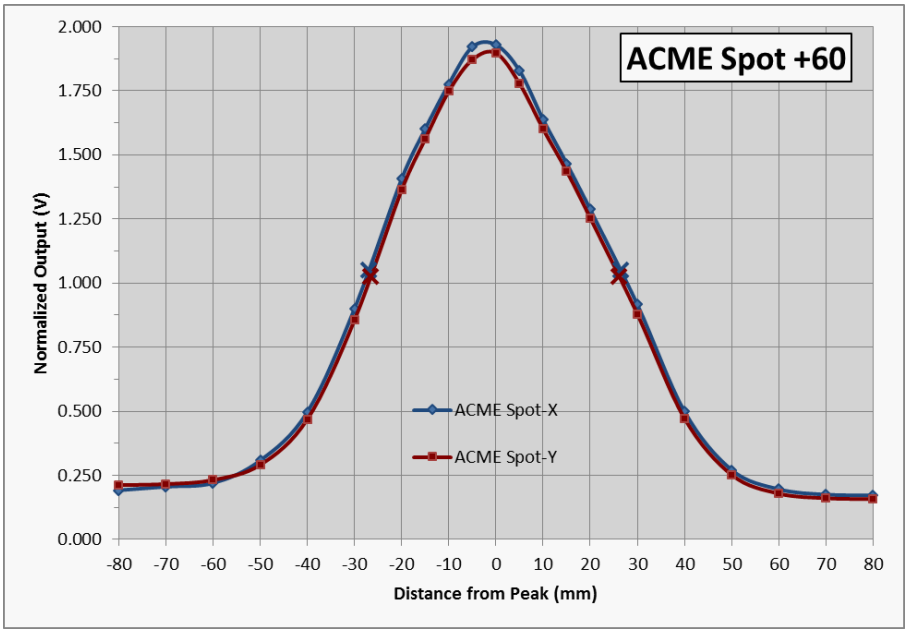


FIGURE 10. +60 mm Spot – FOV Characterization

The following parameter settings were used for the +60 mm Spot FOV scans:

Blackbody Temperature = 950°C
Blackbody Aperture = 0.2”

The following table shows the data resulting from the FOV scans for the +60 mm Spot Radiometer.

TABLE VIII. Data for +60 mm Spot FOV

+60 - X Stats:			+60 - Y Stats:		
P-P	1.757	V	P-P	1.740	V
V_{FWHM}	1.051	V	V_{FWHM}	1.027	V
X1	-27.0	mm	Y1	-26.6	mm
X2	26.4	mm	Y2	26.1	mm
FWHM	53.4	mm	FWHM	52.7	mm
$Center_{FWHM}$	**-0.3**	**mm**	**$Center_{FWHM}$**	**-0.3**	**mm**
$V_{90\%}$	1.753	V	$V_{90\%}$	1.723	V
$X1_{90\%}$	-10.6	mm	$Y1_{90\%}$	-10.7	mm
$X2_{90\%}$	7.0	mm	$Y2_{90\%}$	6.6	mm
$FW_{90\%}$	17.6	mm	$FW_{90\%}$	17.2	mm
$Center_{Alignment}$	**-1.8**	**mm**	**$Center_{Alignment}$**	**-2.1**	**mm**
V_{FOV}	0.348	V	V_{FOV}	0.331	V
$X1_{FOV}$	-47.9	mm	$Y1_{FOV}$	-47.8	mm
$X2_{FOV}$	46.6	mm	$Y2_{FOV}$	46.4	mm
FOV	**94.5**	**mm**	**FOV**	**94.2**	**mm**
$Center_{FOV}$	-0.6	mm	$Center_{FOV}$	-0.7	mm

The results for the +60 mm Spot Radiometer show that the center of symmetry does not completely agree with the geometric peak position, as indicated by the X_{center} and Y_{center} fits of -0.3 mm and -0.3 mm, respectively. These centers were calculated by determining the Full Width at Half Maximum (FWHM) points on each curve and interpolating to get the position of each FWHM point.

The 90% points (10% roll-off) were used to verify the alignment of the detector to the measurement location:

+60 mm Spot Alignment: X = -1.8 mm and Y = -2.1 mm

The 10% points were used to determine the overall Field of View:

+60 mm Spot Radiometer FOV: X = 94.5 mm and Y = 94.2 mm

5.1.6 –60 mm Spot Radiometer Field of View Data / Analysis

Prior to the measurements, the -60 mm Spot detector was aligned to the Blackbody source, by locating the peak of the signal output at the current Gain setting. The resulting detector response to off-axis translations is shown in the figure below. It shows the X (horizontal, along the axis parallel to the line bisecting the detector apertures) and Y (vertical, along the axis perpendicular to the line bisecting the detector apertures) FOV and roll-off measurements.

The following curves show the results from the FOV scans in both the X and Y directions for the -60 mm Spot Radiometer.

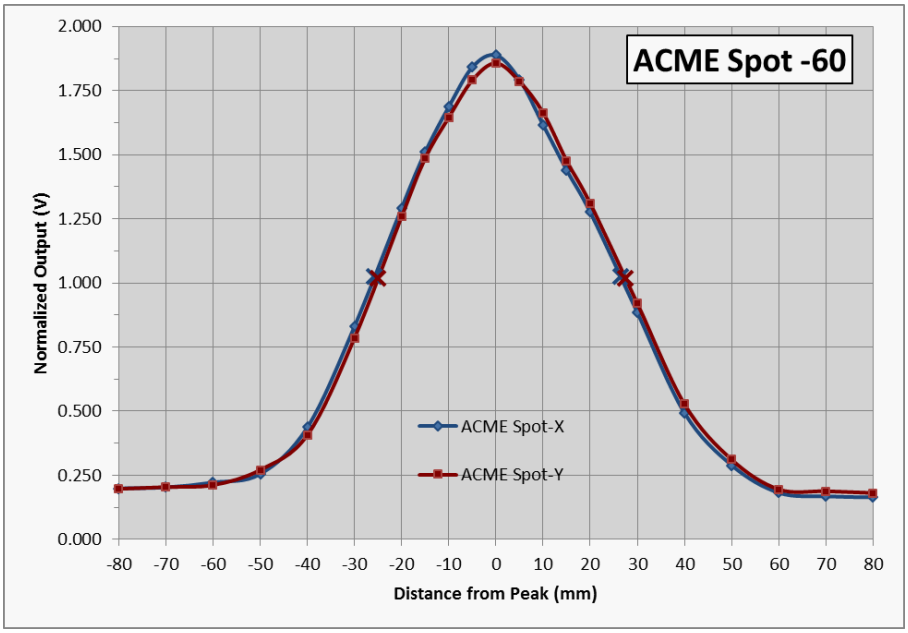


FIGURE 11. –60 mm Spot – FOV Characterization

The following parameter settings were used for the -60 mm Spot FOV scans:

Blackbody Temperature = 900°C
Blackbody Aperture = 0.2”

The following table shows the data resulting from the FOV scans for the -60 mm Spot Radiometer.

TABLE IX. Data for –60 mm Spot FOV

-60 - X Stats:			-60 - Y Stats:		
P-P	1.724	V	P-P	1.676	V
V_{FWHM}	1.026	V	V_{FWHM}	1.019	V
X1	-25.8	mm	Y1	-25.1	mm
X2	26.4	mm	Y2	27.5	mm
FWHM	52.2	mm	FWHM	52.5	mm
$Center_{FWHM}$	**0.3**	**mm**	**$Center_{FWHM}$**	**1.2**	**mm**
$V_{90\%}$	1.716	V	$V_{90\%}$	1.689	V
$X1_{90\%}$	-9.1	mm	$Y1_{90\%}$	-8.5	mm
$X2_{90\%}$	7.2	mm	$Y2_{90\%}$	9.0	mm
$FW_{90\%}$	16.3	mm	$FW_{90\%}$	17.5	mm
$Center_{Alignment}$	**-0.9**	**mm**	**$Center_{Alignment}$**	**0.2**	**mm**
V_{FOV}	0.336	V	V_{FOV}	0.349	V
$X1_{FOV}$	-45.5	mm	$Y1_{FOV}$	-44.3	mm
$X2_{FOV}$	47.7	mm	$Y2_{FOV}$	48.3	mm
FOV	**93.2**	**mm**	**FOV**	**92.6**	**mm**
$Center_{FOV}$	1.1	mm	$Center_{FOV}$	2.0	mm

The results for the -60 mm Spot Radiometer show that the center of symmetry does not completely agree with the geometric peak position, as indicated by the X_{center} and Y_{center} fits of 0.3 mm and 1.2 mm, respectively. These centers were calculated by determining the Full Width at Half Maximum (FWHM) points on each curve and interpolating to get the position of each FWHM point.

The 90% points (10% roll-off) were used to verify the alignment of the detector to the measurement location:

-60 mm Spot Alignment: X = -0.9 mm and Y = 0.2 mm

The 10% points were used to determine the overall Field of View:

-60 mm Spot Radiometer FOV: X = 93.2 mm and Y = 92.6 mm

5.2 ACME Radiometer Assembly Radiometric Calibration

5.2.1 Radiometric Calibration Setup

The Radiometric Calibration is performed separately for each of the 5 detectors in the Radiometer Assembly.

Each detector in the assembly is first centered and aligned normal to the Blackbody source, at the Flight distance of 138.5 mm from the Blackbody Aperture to the outer detector face.

Then for each detector, the Blackbody source was stepped through temperatures ranging from 200°C – 1200°C (373°K - 1473°K). At each temperature, after the Blackbody was allowed to stabilize, the aperture wheel was moved into position in front of the Blackbody output and cycled through aperture sizes of 0.6", 0.4", 0.2", 0.1" and 0.05" with Output Voltage (V_{out}) of the detector recorded at each. This simulates various source Irradiance levels covering the expected operating range. Readings were also taken with the aperture "closed" or covered in order to determine baseline readings for each Radiometer.

The Table below shows the expected Irradiance Range matrix for the calibration of the Radiometers.

TABLE X. Expected Irradiance Range for ACME Radiometer Assembly

Temp. (°C)	Aperture Diameter 0.6	0.4	0.2	0.1	0.05
200	10% 0.00105			Min 0.000024	
300					
400					Min 0.000024
460	50% 0.00525		5% 0.000525		
500					
600	Max 0.0105		10% 0.00105		
650		50% 0.00525			
700					
800		Max 0.0105		5% 0.000525	
900					
950				10% 0.00105	
1000			50% 0.00525		
1100					
1200			Max 0.0105		5% 0.000525

5.2.2 Center – Wide Angle Radiometric Calibration Data / Analysis

Using the setup and method described above, the Radiometric Calibration data were collected for the Center – Wide Angle detector and are shown in the Table below. The Output Voltages (V_{out}) recorded correspond to Irradiance values (W/cm^2) that are defined by the fixed distance from the Blackbody source and the temperature – aperture diameter combination. The grayed out boxes indicate the readings were out of range, either on the low or high end, and have been excluded from the measurements and plots.

TABLE XI. Center – Wide Angle Radiometric Calibration Data

C	Aperture = 0.6"		Aperture = 0.4"		Aperture = 0.2"		Aperture = 0.1"		Aperture = 0.05"		Covered	
T (°C)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	T (°C)	V_{out}
200	1.651	0.0008602	1.501	0.0003823	1.361	0.0000956	1.314	0.0000239			200	1.225
300	2.004	0.0018522	1.644	0.0008232	1.439	0.0002058	1.345	0.0000515			300	1.221
400	2.513	0.0035242	1.927	0.0015663	1.506	0.0003916	1.367	0.0000979	1.319	0.0000245	400	1.218
460	2.985	0.0049589	2.118	0.0022040	1.559	0.0005510	1.396	0.0001377	1.331	0.0000344	460	1.211
500	3.404	0.0061330	2.316	0.0027258	1.621	0.0006814	1.432	0.0001704	1.339	0.0000426	500	1.198
600	4.761	0.0099764	2.903	0.0044339	1.762	0.0011085	1.457	0.0002771	1.353	0.0000693	600	1.203
650	4.991	0.0124654	3.253	0.0055402	1.872	0.0013850	1.487	0.0003463	1.358	0.0000866	650	1.182
700	(Saturated)		3.798	0.0068415	1.981	0.0017104	1.515	0.0004276	1.379	0.0001069	700	1.179
800			4.828	0.0101176	2.265	0.0025294	1.618	0.0006323	1.423	0.0001581	800	1.146
900			4.991	0.0144493	2.624	0.0036123	1.706	0.0009031	1.445	0.0002258	900	1.125
950			(Saturated)		2.827	0.0042687	1.748	0.0010672	1.449	0.0002668	950	1.143
1000					3.125	0.0050106	1.829	0.0012527	1.472	0.0003132	1000	1.096
1100					3.767	0.0067803	1.965	0.0016951	1.512	0.0004238	1100	1.017
1200					4.468	0.0089818	2.129	0.0022454	1.566	0.0005614	1200	0.792
											Avg =	1.140

The above data show the voltage range covers ~ 3.85 V, with the maximum output voltage, just below saturation of V_{max} = 4.99 V and the average minimum voltage, V_{min} = 1.140 V.

The measured Output Voltage range of 1.31 V – 4.99 V corresponds to an overall Irradiance Range of:

2.39×10^{-5} W/cm^2 to 1.05×10^{-2} W/cm^2

The data shown in the above table has been graphed independent of the aperture/temperature combination in order to produce the corresponding overall Irradiance values. The Figure below shows the overlay of the linear data fit correlation between Output Voltage and Irradiance by method of least squares. The corresponding equation with slope, intercept and correlation coefficient are also shown.

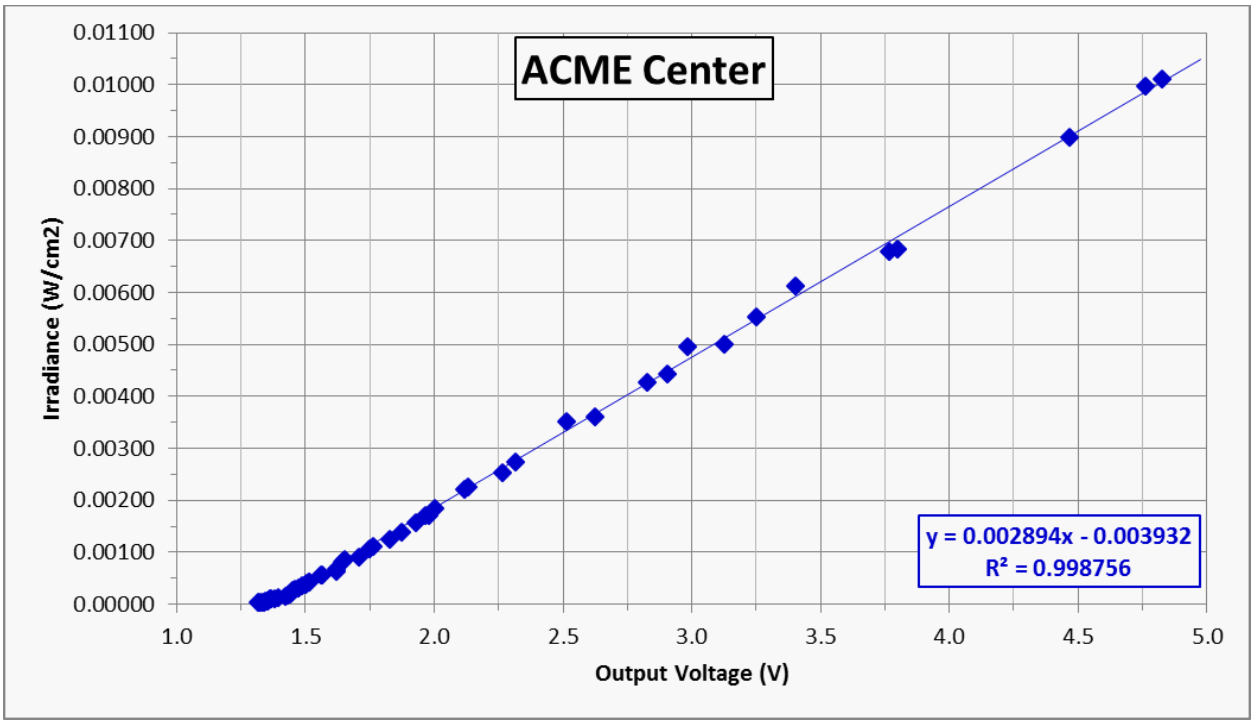


FIGURE 12. Center – Wide Angle Radiometric Data with Linear Fit

5.2.3 +30 mm – Spot Radiometric Calibration Data / Analysis

Using the setup and method described above, the Radiometric Calibration data were collected for the +30 mm Spot detector and are shown in the Table below. The Output Voltages (V_{out}) recorded correspond to Irradiance values (W/cm^2) that are defined by the fixed distance from the Blackbody source and the temperature – aperture diameter combination. The grayed out boxes indicate the readings were out of range, either on the low or high end, and have been excluded from the measurements and plots.

TABLE XII. +30 mm – Spot Radiometric Calibration Data

+30	Aperture = 0.6"		Aperture = 0.4"		Aperture = 0.2"		Aperture = 0.1"		Aperture = 0.05"		Covered	
T (°C)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	T (°C)	V_{out}
200	1.711	0.00086022	1.504	0.0003823	1.354	0.0000956	1.291	0.0000239			200	1.232
300	2.229	0.00185221	1.697	0.0008232	1.415	0.0002058	1.329	0.0000515	1.272	0.0000129	300	1.228
400	3.173	0.00352423	2.151	0.00156632	1.533	0.0003916	1.357	0.0000979	1.292	0.0000245	400	1.231
460	3.996	0.00495893	2.544	0.0022040	1.593	0.0005510	1.373	0.0001377	1.303	0.0000344	460	1.223
500	4.636	0.0061330	2.849	0.00272577	1.638	0.0006814	1.396	0.0001704	1.315	0.0000426	500	1.217
600	4.991	0.00997635	3.792	0.0044339	1.889	0.0011085	1.472	0.0002771	1.332	0.0000693	600	1.221
650	(Saturated)		4.425	0.0055402	2.007	0.0013850	1.498	0.0003463	1.344	0.0000866	650	1.213
700			4.991	0.0068415	2.212	0.0017104	1.553	0.0004276	1.367	0.0001069	700	1.211
750			(Saturated)		2.438	0.0020899	1.581	0.00052248	1.369	0.0001306	750	1.207
800					2.681	0.0025294	1.624	0.00063235	1.388	0.0001581	800	1.203
850					2.988	0.0030348	1.682	0.00075869	1.410	0.0001897	850	1.201
900					3.314	0.0036123	1.781	0.00090308	1.426	0.0002258	900	1.204
1000					4.183	0.0050106	1.954	0.00125266	1.487	0.0003132	1000	1.198
1100					4.991	0.0067803	2.195	0.00169506	1.552	0.0004238	1100	1.205
1200					(Saturated)		2.626	0.0022454	1.616	0.0005614	1200	1.209
											Avg =	1.214

The above data show the voltage range covers ~ 3.78 V, with the maximum output voltage, just below saturation of V_{max} = 4.99 V and the average minimum voltage, V_{min} = 1.214 V.

The measured Output Voltage range of 1.27 V – 4.99 V corresponds to an overall Irradiance Range of:

1.29 x 10^{-5} W/cm^2 to 6.64 x 10^{-3} W/cm^2

The data shown in the above table has been graphed independent of the aperture/temperature combination in order to produce the corresponding overall Irradiance values. The Figure below shows the overlay of the linear data fit correlation between Output Voltage and Irradiance by method of least squares. The corresponding equation with slope, intercept and correlation coefficient are also shown.

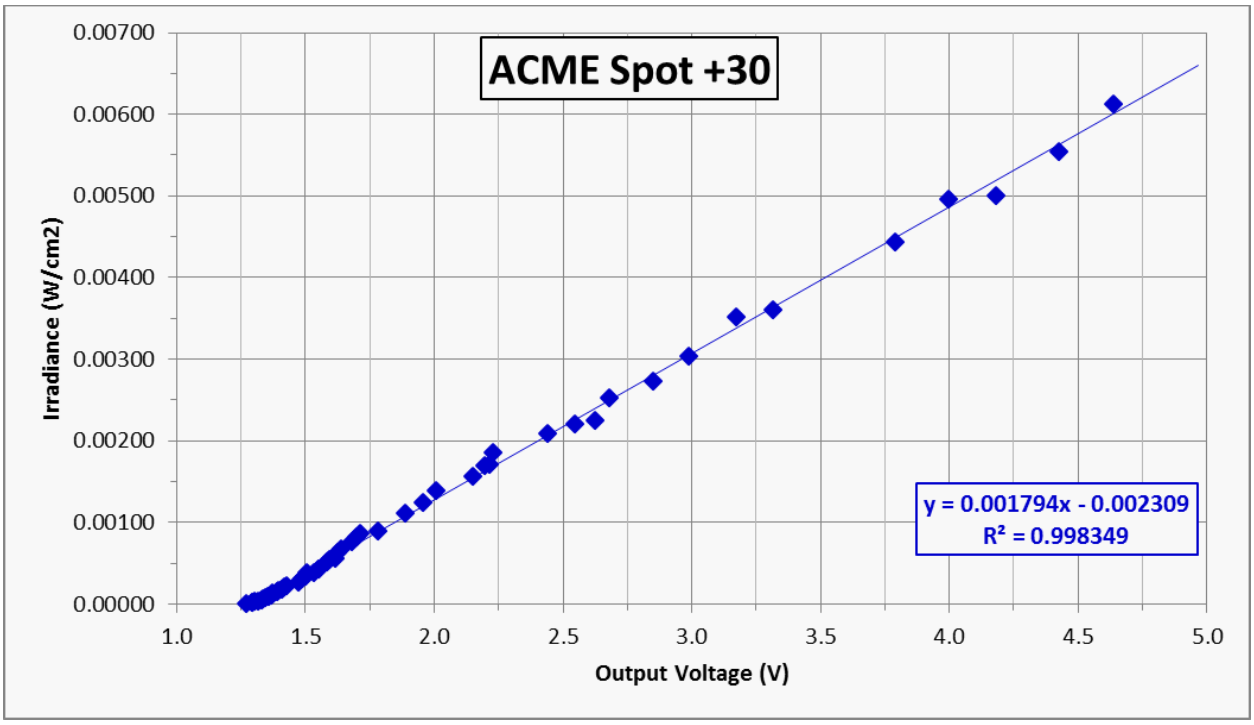


FIGURE 13. +30 mm – Spot Radiometric Data with Linear Fit

5.2.4 –30 mm Spot Radiometric Calibration Data / Analysis

Using the setup and method described above, the Radiometric Calibration data were collected for the –30 mm Spot detector and are shown in the Table below. The Output Voltages (V_{out}) recorded correspond to Irradiance values (W/cm^2) that are defined by the fixed distance from the Blackbody source and the temperature – aperture diameter combination. The grayed out boxes indicate the readings were out of range, either on the low or high end, and have been excluded from the measurements and plots.

TABLE XIII. –30 mm Spot Radiometric Calibration Data

-30	Aperture = 0.6"		Aperture = 0.4"		Aperture = 0.2"		Aperture = 0.1"		Aperture = 0.05"		Covered	
T (°C)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	T (°C)	V_{out}
200	1.617	0.00086022	1.431	0.0003823	1.336	0.0000956	1.289	0.0000239			200	1.226
300	2.113	0.00185221	1.593	0.0008232	1.387	0.0002058	1.312	0.0000515	1.277	0.0000129	300	1.224
400	2.645	0.00352423	1.853	0.00156632	1.439	0.0003916	1.337	0.0000979	1.296	0.0000245	400	1.229
500	3.788	0.0061330	2.344	0.00272577	1.553	0.0006814	1.381	0.0001704	1.306	0.0000426	500	1.224
550	4.618	0.00788011	2.622	0.0035023	1.633	0.0008756	1.398	0.0002189	1.315	0.0000547	550	1.219
600	4.992	0.00997635	3.083	0.0044339	1.747	0.0011085	1.419	0.0002771	1.323	0.0000693	600	1.214
700	(Saturated)		4.214	0.0068415	2.016	0.0017104	1.497	0.0004276	1.344	0.0001069	700	1.208
800			4.992	0.0101176	2.329	0.0025294	1.549	0.00063235	1.365	0.0001581	800	1.154
900			(Saturated)		2.796	0.0036123	1.682	0.00090308	1.415	0.0002258	900	1.147
950					3.129	0.0042687	1.764	0.00106717	1.417	0.0002668	950	1.147
1000					3.481	0.0050106	1.809	0.00125266	1.423	0.0003132	1000	1.134
1100					4.151	0.0067803	1.961	0.00169506	1.495	0.0004238	1100	1.145
1200					4.992	0.0089818	2.252	0.0022454	1.533	0.0005614	1200	1.176
											Avg =	1.188

The above data show the voltage range covers ~ 3.80 V, with the maximum output voltage, just below saturation of V_{max} = 4.99 V and the average minimum voltage, V_{min} = 1.188 V.

The measured Output Voltage range of 1.28 V – 4.99 V corresponds to an overall Irradiance Range of:

1.29×10^{-5} W/cm^2 to 8.88×10^{-3} W/cm^2

The data shown in the above table has been graphed independent of the aperture/temperature combination in order to produce the corresponding overall Irradiance values. The Figure below shows the overlay of the linear data fit correlation between Output Voltage and Irradiance by method of least squares. The corresponding equation with slope, intercept and correlation coefficient are also shown.

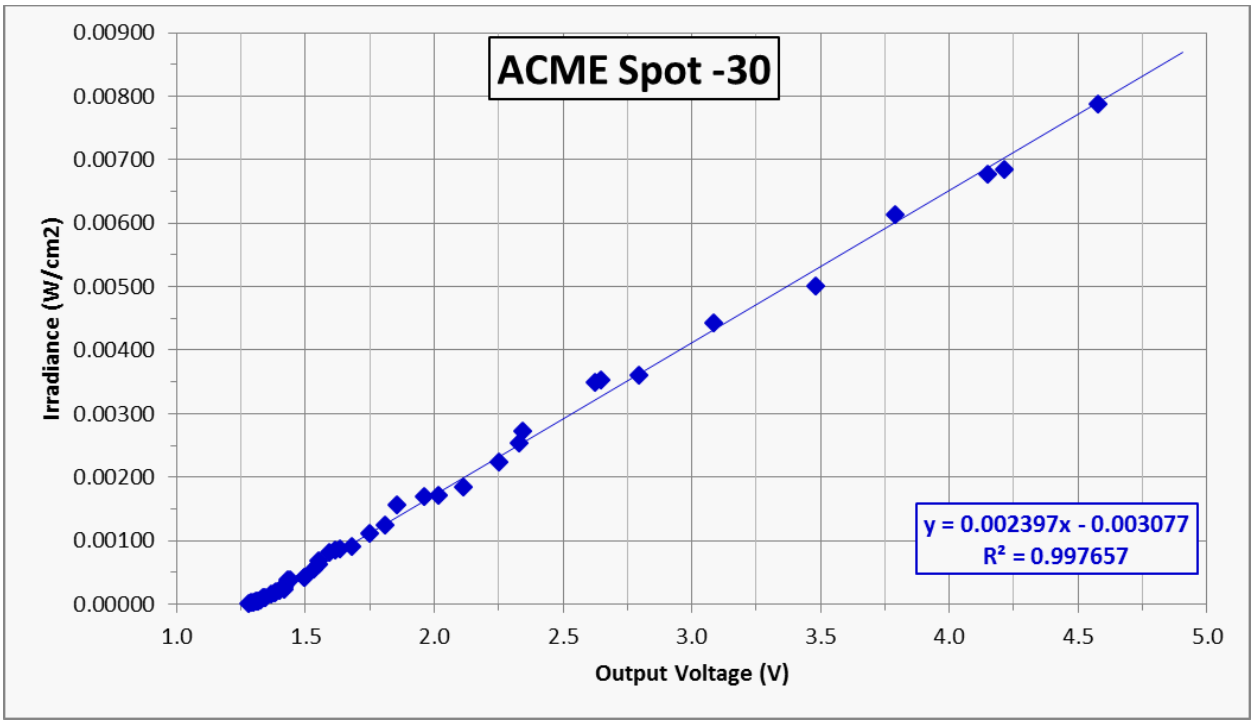


FIGURE 14. – 30 mm Spot Radiometric Data with Linear Fit

5.2.5 +60 mm Spot Radiometric Calibration Data / Analysis

Using the setup and method described above, the Radiometric Calibration data were collected for the +60 mm Spot detector and are shown in the Table below. The Output Voltages (V_{out}) recorded correspond to Irradiance values (W/cm^2) that are defined by the fixed distance from the Blackbody source and the temperature – aperture diameter combination. The grayed out boxes indicate the readings were out of range, either on the low or high end, and have been excluded from the measurements and plots.

TABLE XIV. +60 mm Spot Radiometric Calibration Data

+60	Aperture = 0.6"		Aperture = 0.4"		Aperture = 0.2"		Aperture = 0.1"		Aperture = 0.05"		Covered	
T (°C)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	T (°C)	V_{out}
200	1.569	0.00086022	1.417	0.0003823	1.326	0.0000956	1.289	0.0000239			200	1.116
300	2.012	0.00185221	1.554	0.0008232	1.360	0.0002058	1.309	0.0000515	1.286	0.0000129	300	1.113
400	2.532	0.00352423	1.805	0.00156632	1.423	0.0003916	1.339	0.0000979	1.297	0.0000245	400	1.115
500	3.465	0.0061330	2.268	0.00272577	1.523	0.0006814	1.352	0.0001704	1.308	0.0000426	500	1.121
550	4.182	0.00788011	2.516	0.0035023	1.629	0.0008756	1.376	0.0002189	1.315	0.0000547	550	1.125
600	4.991	0.00997635	3.010	0.0044339	1.737	0.0011085	1.405	0.0002771	1.318	0.0000693	600	1.122
700	(Saturated)		**3.988**	0.0068415	1.952	0.0017104	1.462	0.0004276	1.340	0.0001069	700	1.124
800			4.991	0.0101176	2.261	0.0025294	1.498	0.00063235	1.351	0.0001581	800	1.113
900			(Saturated)		2.724	0.0036123	1.645	0.00090308	1.382	0.0002258	900	1.111
950					2.989	0.0042687	1.731	0.00106717	1.398	0.0002668	950	1.106
1000					3.327	0.0050106	1.793	0.00125266	1.409	0.0003132	1000	1.118
1100					**3.916**	0.0067803	1.936	0.00169506	1.457	0.0004238	1100	1.126
1200					4.991	0.0089818	2.192	0.0022454	1.494	0.0005614	1200	1.133
											Avg =	1.119

The above data show the voltage range covers ~ 3.87 V, with the maximum output voltage, just below saturation of V_{max} = 4.99 V and the average minimum voltage, V_{min} = 1.119 V.

The measured Output Voltage range of 1.286 V – 4.99 V corresponds to an overall Irradiance Range of:

1.29×10^{-5} W/cm^2 to 9.71×10^{-3} W/cm^2

The data shown in the above table has been graphed independent of the aperture/temperature combination in order to produce the corresponding overall Irradiance values. The Figure below shows the overlay of the linear data fit correlation between Output Voltage and Irradiance by method of least squares. The corresponding equation with slope, intercept and correlation coefficient are also shown.

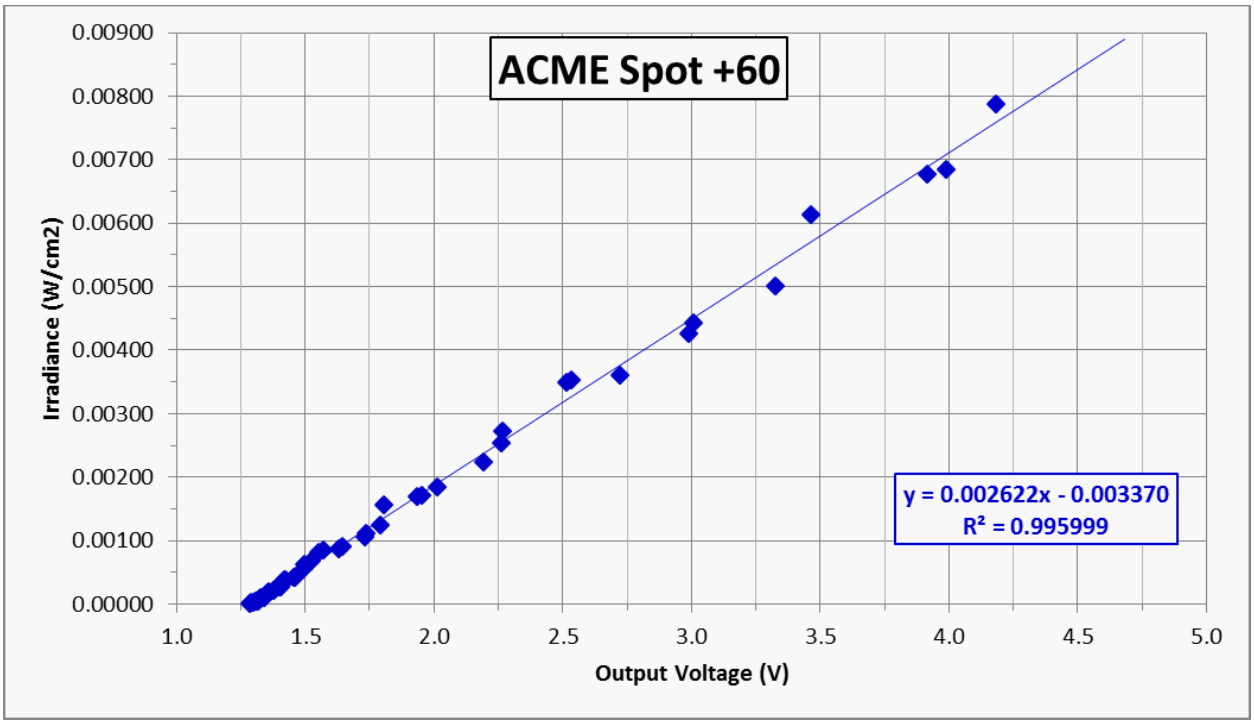


FIGURE 15. +60 mm Spot Radiometric Data with Linear Fit

5.2.6 –60 mm Spot Radiometric Calibration Data / Analysis

Using the setup and method described above, the Radiometric Calibration data were collected for the –60 mm Spot detector and are shown in the Table below. The Output Voltages (V_{out}) recorded correspond to Irradiance values (W/cm^2) that are defined by the fixed distance from the Blackbody source and the temperature – aperture diameter combination. The grayed out boxes indicate the readings were out of range, either on the low or high end, and have been excluded from the measurements and plots.

TABLE XV. –60 mm Spot Radiometric Calibration Data

-60	Aperture = 0.6"		Aperture = 0.4"		Aperture = 0.2"		Aperture = 0.1"		Aperture = 0.05"		Covered	
T (°C)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	T (°C)	V_{out}
200	1.553	0.00086022	1.398	0.0003823	1.312	0.0000956	1.285	0.0000239			200	1.225
300	1.987	0.00185221	1.543	0.0008232	1.353	0.0002058	1.295	0.0000515	1.274	0.0000129	300	1.226
400	2.654	0.00352423	1.781	0.00156632	1.416	0.0003916	1.321	0.0000979	1.286	0.0000245	400	1.225
500	3.481	0.0061330	2.334	0.00272577	1.534	0.0006814	1.341	0.0001704	1.291	0.0000426	500	1.221
550	4.173	0.00788011	2.594	0.0035023	1.630	0.0008756	1.362	0.0002189	1.304	0.0000547	550	1.222
600	4.992	0.00997635	2.997	0.0044339	1.727	0.0011085	1.375	0.0002771	1.307	0.0000693	600	1.217
700	(Saturated)		4.011	0.0068415	1.965	0.0017104	1.452	0.0004276	1.324	0.0001069	700	1.214
800			4.992	0.0101176	2.319	0.0025294	1.521	0.00063235	1.335	0.0001581	800	1.193
900			(Saturated)		2.711	0.0036123	1.635	0.00090308	1.368	0.0002258	900	1.187
950					2.972	0.0042687	1.703	0.00106717	1.371	0.0002668	950	1.179
1000					3.236	0.0050106	1.767	0.00125266	1.391	0.0003132	1000	1.159
1100					3.971	0.0067803	1.907	0.00169506	1.432	0.0004238	1100	1.156
1200					4.992	0.0089818	2.171	0.0022454	1.474	0.0005614	1200	1.168
											Avg =	1.199

The above data show the voltage range covers ~ 3.79 V, with the maximum output voltage, just below saturation of V_{max} = 4.99 V and the average minimum voltage, V_{min} = 1.199 V.

The measured Output Voltage range of 1.27 V – 4.99 V corresponds to an overall Irradiance Range of:

1.29×10^{-5} W/cm^2 to 9.62×10^{-3} W/cm^2

The data shown in the above table has been graphed independent of the aperture/temperature combination in order to produce the corresponding overall Irradiance values. The Figure below shows the overlay of the linear data fit correlation between Output Voltage and Irradiance by method of least squares. The corresponding equation with slope, intercept and correlation coefficient are also shown.

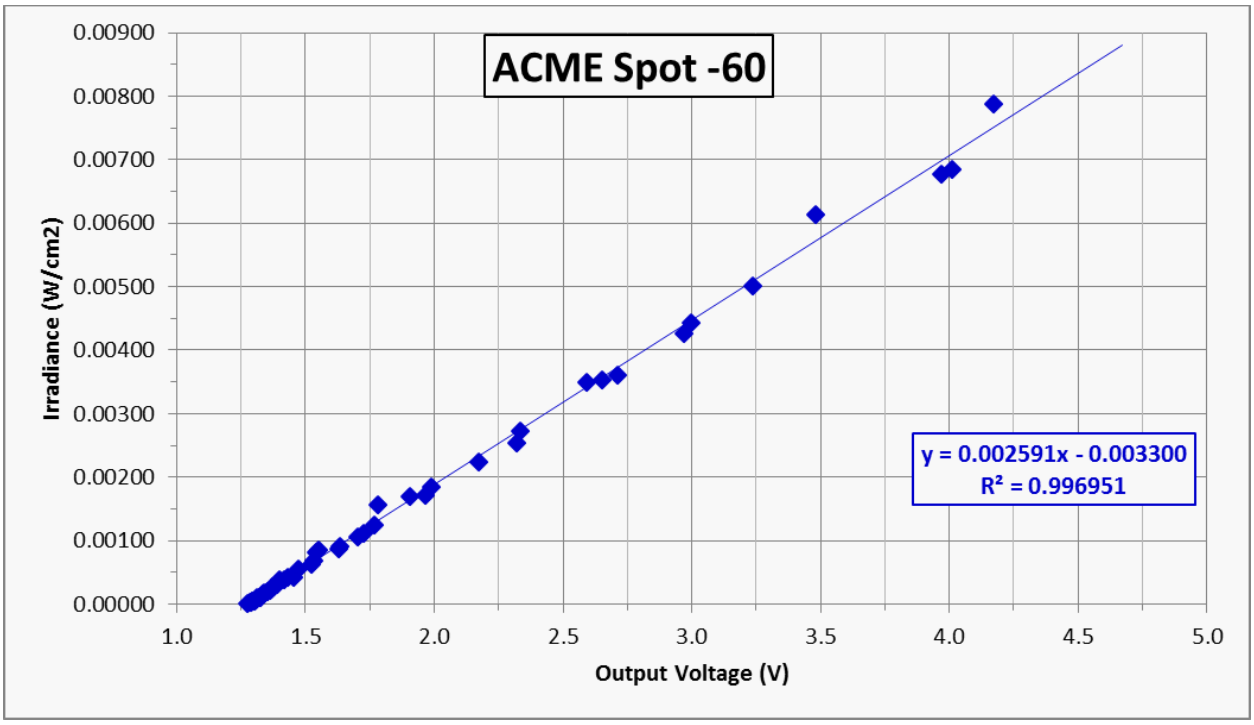


FIGURE 16. –60 mm Spot Radiometric Data with Linear Fit

5.3 Response Time

5.3.1 Response Time Analysis

The response time of each of the thermopile detectors in the ACME Radiometer Assembly was measured using an optical chopper wheel to provide a step function as irradiance input. The Output Voltage of each detector is then analyzed to calculate the time to rise from 0 – 62.5% of the signal range.

The optical chopper and controller used are ThorLabs Model MC1000. The wheel was configured for 2 cycles (i.e. 2 open regions and 2 blocked regions); hence a 1 Hz drive produces a 2 Hz response. The Figure below shows the results of the time response testing for all 5 of the detectors in the ACME Radiometer Assembly.

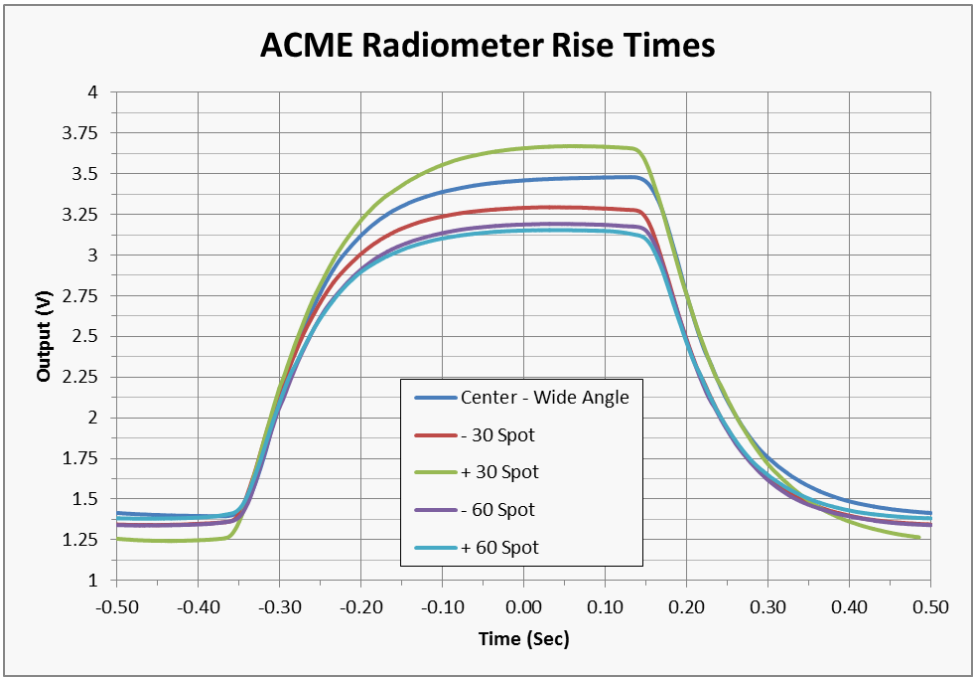


FIGURE 17. ACME Radiometer Assembly Time Response

The table below shows the calculated response time for each detector based on the method above.

TABLE XVI. ACME Radiometer Assembly Response Time Values

	Center		+30 Spot		-30 Spot		+60 Spot		-60 Spot	
	V_{Out}	Time (s)	V_{Out}	Time (s)	V_{Out}	Time (s)	V_{Out}	Time (s)	V_{Out}	Time (s)
Min V	1.39	-0.35890	1.39	-0.41760	1.34	-0.39640	1.38	-0.44390	1.34	-0.37190
Max V	3.48		4.11		3.29		3.15		3.19	
Delta V	2.09		2.72		1.95		1.78		1.85	
62.50%	2.18	-0.28200	2.41	-0.28790	2.07	-0.25690	2.04	-0.30380	2.03	-0.23330
Time Delta		0.07690		0.12970		0.13950		0.14010		0.13860
Response Time		**76.9 ms**		**129.7 ms**		**139.5 ms**		**140.1 ms**		**138.6 ms**

5.4 Statistics & Error Analysis

The accuracy is determined by a combination of random and systematic errors accounted for in the accompanying statistical analysis. This analysis represents the graphical data by method of least squares. Note the calculated Irradiances in the plots are all subject to the uncertainties and the Output Voltages recorded in the calibration also have measurement uncertainties.

The graphical approach used does not inherently account for unknown systematic errors.

The Table below summarizes the results of the graphical analysis of the collected data for all 5 of the detectors in the ACME Radiometer Assembly.

TABLE XVII. ACME Radiometer Assembly Statistical Analysis

Radiometer Statistics	Center	+30	-30	+60	-60
Voltage Range (V)	3.51	3.36	3.30	2.90	2.90
Min Voltage (V)	1.31	1.27	1.28	1.29	1.27
Max Voltage (V)	4.83	4.64	4.58	4.18	4.17
Irradiance Range (W/cm^2)	0.0100937	0.0061201	0.0078672	0.0078672	0.0078672
Min Irradiance (W/cm^2)	0.0000239	0.0000129	0.0000129	0.0000129	0.0000129
Max Irradiance (W/cm^2)	0.0101176	0.0061330	0.0078801	0.0078801	0.0078801
Uncertainty (± V)	1.4032E-05	1.0118E-05	1.7510E-05	2.4236E-05	2.0900E-05
(Linear) R^2	**0.9988**	**0.9983**	**0.9977**	**0.9960**	**0.9970**
(Linear) Slope	0.002894	0.001794	0.002397	0.002622	0.002591
(Linear) Intercept	-0.003932	-0.002309	-0.003077	-0.003370	-0.003300

The Table below shows the final Total Error at the Minimum Irradiance and Maximum Irradiance values, as well as at values representing 5%, 10% and 50% of the Max value.

In addition, the Irradiance at the 5% Error threshold has been calculated and is displayed.

TABLE XVIII. ACME Radiometer Assembly Accuracy & Error Analysis

Accuracy/Error Summary	Center	+30	-30	+60	-60
Error at Min	13.36%	9.64%	16.68%	23.08%	19.91%
Error at 5% Max	2.67%	1.93%	3.34%	4.62%	3.98%
Error at 10% Max	1.34%	0.96%	1.67%	2.31%	1.99%
Error at ½ Max	0.27%	0.19%	0.33%	0.46%	0.40%
Error at Max	0.13%	0.10%	0.17%	0.23%	0.20%
Min Irradiance @ 5% Error	0.0002806	0.0002024	0.0003502	0.0004847	0.0004180

6.0 60º RADIOMETER ASSEMBLY CHARACTERIZATION

6.1 60 Degree Offset Radiometer Assembly Field of View

6.1.1 FOV Setup

The Field of View (FOV) measurement was performed for the single thermopile detector in the 60 Degree Offset Radiometer Assembly. The single detector is centered and aligned normal to the Blackbody source, at a distance of 157.5 mm from the Blackbody Aperture to the outer face. The detector is then translated in the horizontal (X) axis over a range of ±70 mm about the center position, in 10 mm increments, while recording the output signal level. This determines the X axis FOV and roll-off characteristics.

While positioned at the X axis center point, the vertical (Y) axis characterization is measured by raising/lowering the assembly on a lab jack over ±70 mm to measure the Y axis FOV and roll-off characteristics.

6.1.2 60 Degree Offset Radiometer Field of View Data / Analysis

Prior to the measurements, the 60 Degree Offset detector was aligned to the Blackbody source, by locating the peak of the signal output at the current Gain setting. The resulting detector response to off-axis translations is shown in the figure below. It shows the X (horizontal, along the axis parallel to the line bisecting the detector apertures) and Y (vertical, along the axis perpendicular to the line bisecting the detector apertures) FOV and roll-off measurements.

The following curves show the results from the FOV scans in both the X and Y directions for the 60 Degree Offset Radiometer.

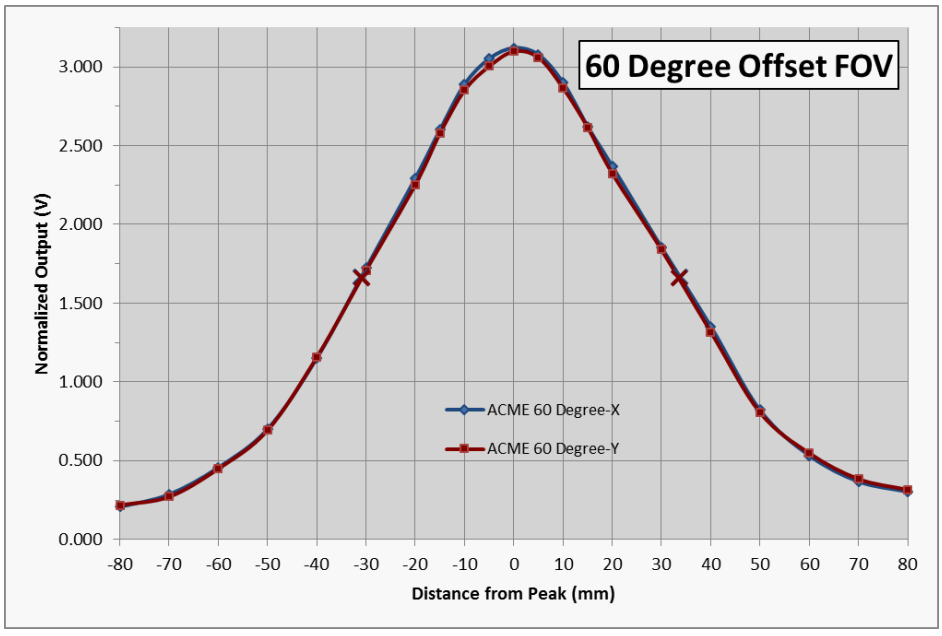


FIGURE 18. 60 Degree Offset – FOV Characterization

The following parameter settings were used for the 60 Degree Offset FOV scans:

Blackbody Temperature = 1000°C
Blackbody Aperture = 0.25”

The following table shows the data resulting from the FOV scans for the 60 Degree Offset Radiometer.

TABLE XIX. Data for 60 Degree Offset FOV

60 Degree - X Stats:			60 Degree - Y Stats:		
P-P	2.913	V	P-P	2.879	V
V_{FWHM}	1.664	V	V_{FWHM}	1.658	V
X1	-31.0	mm	Y1	-30.9	mm
X2	33.8	mm	Y2	33.5	mm
FWHM	64.8	mm	FWHM	64.4	mm
$Center_{FWHM}$	**1.4**	**mm**	**$Center_{FWHM}$**	**1.3**	**mm**
$V_{90\%}$	2.829	V	$V_{90\%}$	2.809	V
$X1_{90\%}$	-11.0	mm	$Y1_{90\%}$	-10.8	mm
$X2_{90\%}$	11.3	mm	$Y2_{90\%}$	11.2	mm
$FW_{90\%}$	22.3	mm	$FW_{90\%}$	21.9	mm
$Center_{Alignment}$	**0.1**	**mm**	**$Center_{Alignment}$**	**0.2**	**mm**
V_{FOV}	0.498	V	V_{FOV}	0.506	V
$X1_{FOV}$	-58.3	mm	$Y1_{FOV}$	-57.7	mm
$X2_{FOV}$	62.1	mm	$Y2_{FOV}$	62.6	mm
FOV	**120.4**	**mm**	**FOV**	**120.2**	**mm**
$Center_{FOV}$	1.9	mm	$Center_{FOV}$	2.4	mm

The results for the 60 Degree Offset Radiometer show that the center of symmetry does not completely agree with the geometric peak position, as indicated by the X_{center} and Y_{center} fits of -1.4 mm and 1.3 mm, respectively. These centers were calculated by determining the Full Width at Half Maximum (FWHM) points on each curve and interpolating to get the position of each FWHM point.

The 90% points (10% roll-off) were used to verify the alignment of the detector to the measurement location:

60 Degree Offset Alignment: X = 0.1 mm and Y = 0.2 mm

The 10% points were used to determine the overall Field of View:

60 Degree Offset Radiometer FOV: X = 120.4 mm and Y = 120.2 mm

6.2 60 Degree Radiometer Assembly Radiometric Calibration

6.2.1 Radiometric Calibration Setup

The Radiometric Calibration was performed for the single thermopile detector in the 60 Degree Offset Radiometer Assembly.

The single detector is first centered and aligned normal to the Blackbody source, at the Flight distance of 157.5 mm from the Blackbody Aperture to the outer detector face.

Then the Blackbody source was stepped through temperatures ranging from 200°C – 1200°C (373°K - 1473°K). At each temperature, after the Blackbody was allowed to stabilize, the aperture wheel was moved into position in front of the Blackbody output and cycled through aperture sizes of 0.25", 0.2", 0.1" and 0.05" with Output Voltage (V_{out}) of the detector recorded at each. This simulates various source Irradiance levels covering the expected operating range. Readings were also taken with the aperture "closed" or covered in order to determine baseline readings for each Radiometer.

The Table below shows the expected Irradiance Range matrix for the calibration of this Radiometer.

TABLE XX. Expected Irradiance Range for 60 Degree Offset

Temp. (°C)	Aperture Diameter 0.25	0.2	0.1	0.05
200			Min 0.0000185	
300				
400		5% 0.00032		
500				Min 0.0000185
600				
700			5% 0.00032	
800	50% 0.0032			
900			10% 0.00064	
950		50% 0.0032		
1000	Max 0.0064			
1050				
1100				
1150		Max 0.0064		5% 0.00032
1200				

6.2.2 60 Degree Offset Radiometric Calibration Data / Analysis

Using the setup and method described above, the Radiometric Calibration data were collected for the 60 Degree Offset detector and are shown in the Table below. The Output Voltages (V_{out}) recorded correspond to Irradiance values (W/cm^2) that are defined by the fixed distance from the Blackbody source and the temperature – aperture diameter combination. The grayed out boxes indicate the readings were out of range, either on the low or high end, and have been excluded from the measurements and plots.

TABLE XXI. 60 Degree Offset Radiometric Calibration Data

Offset	Aperture = 0.25"		Aperture = 0.2"		Aperture = 0.1"		Aperture = 0.05"		Covered	
T (°C)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	V_{out}	G (W/cm^2)	T (°C)	V_{out}
200	1.33	0.0001155	1.31	0.0000739	1.27	0.0000185			200	1.16
300	1.41	0.0002487	1.37	0.0001591	1.29	0.0000398			300	1.19
400	1.53	0.0004731	1.45	0.0003028	1.32	0.0000757			400	1.18
500	1.71	0.0008234	1.59	0.0005269	1.36	0.0001317	1.28	0.0000329	500	1.16
600	2.05	0.0013393	1.77	0.0008572	1.39	0.0002143	1.30	0.0000536	600	1.17
700	2.43	0.0020666	2.04	0.0013226	1.47	0.0003307	1.32	0.0000827	700	1.19
750	2.68	0.0025252	2.21	0.0016161	1.51	0.0004040	1.33	0.0001010	750	1.20
800	2.96	0.0030561	2.39	0.0019559	1.58	0.0004890	1.34	0.0001222	800	1.20
850	3.31	0.0036668	2.61	0.0023467	1.64	0.0005867	1.36	0.0001467	850	1.19
900	3.67	0.0043646	2.85	0.0027934	1.69	0.0006983	1.38	0.0001746	900	1.20
950	4.12	0.0051576	3.14	0.0033009	1.74	0.0008252	1.39	0.0002063	950	1.21
1000	4.65	0.0060541	3.46	0.0038746	1.84	0.0009687	1.41	0.0002422	1000	1.20
1050	4.88	0.0070627	3.81	0.0045201	1.93	0.0011300	1.44	0.0002825	1050	1.21
1100	(Saturated)		4.21	0.0052431	2.02	0.0013108	1.46	0.0003277	1100	1.21
1150			4.61	0.0060494	2.16	0.0015124	1.50	0.0003781	1150	1.21
1200			4.88	0.0069454	2.29	0.0017364	1.52	0.0004341	1200	1.22
									Avg =	1.194

The above data show the voltage range covers ~ 3.69 V, with the maximum output voltage, just below saturation of V_{max} = 4.88 V and the average minimum voltage, V_{min} = 1.194 V.

The measured Output Voltage range of 1.27 V – 4.88 V corresponds to an overall Irradiance Range of:

1.85×10^{-5} W/cm^2 to 6.47×10^{-3} W/cm^2

The data shown in the above table has been graphed independent of the aperture/temperature combination in order to produce the corresponding overall Irradiance values. The Figure below shows the overlay of the linear data fit correlation between Output Voltage and Irradiance by method of least squares. The corresponding equation with slope, intercept and correlation coefficient are also shown.

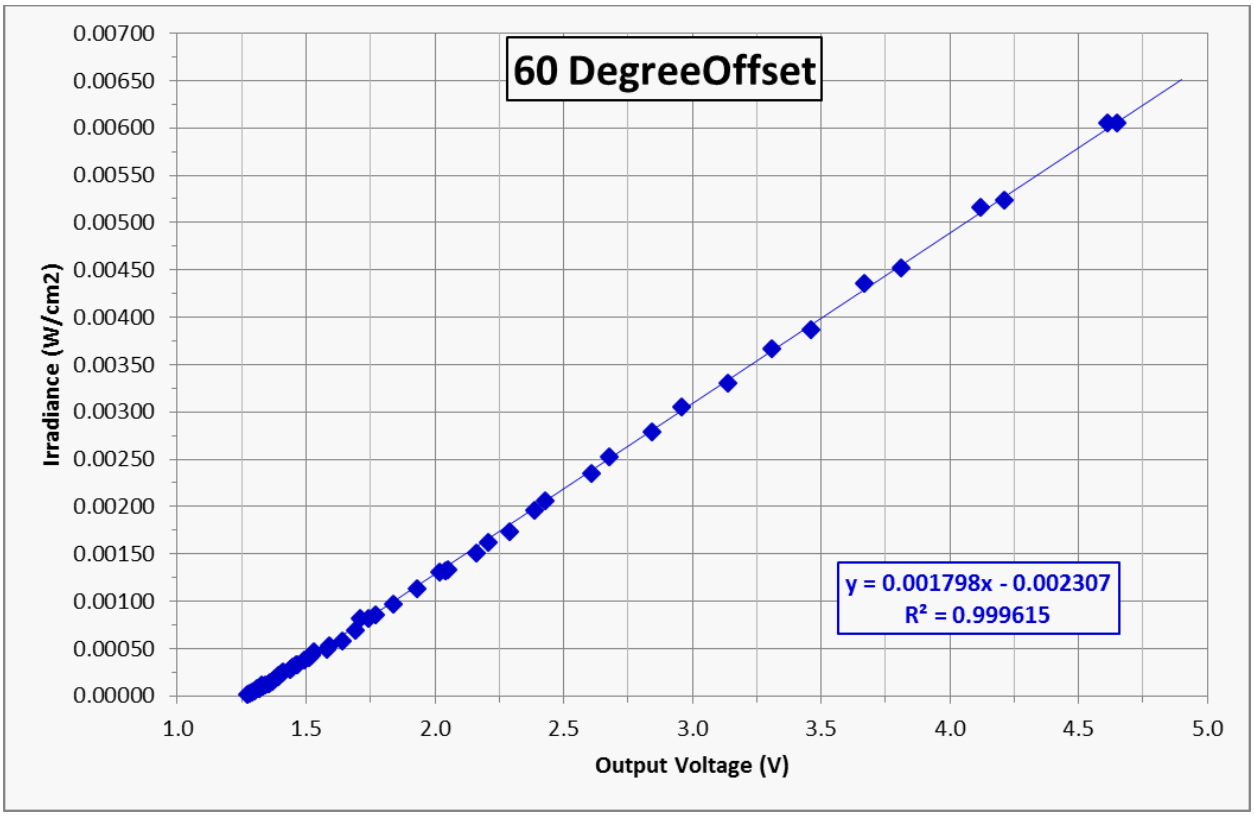


FIGURE 19. 60 Degree Offset Radiometric Data with Linear Fit

6.3 60 Degree Offset Radiometer Response Time

6.3.1 Response Time Analysis

The response time of the thermopile detector in the 60 Degree Offset Radiometer Assembly was measured using an optical chopper wheel to provide a step function as irradiance input. The Output Voltage of each detector is then analyzed to calculate the time to rise from 0 – 62.5% of the signal range.

The optical chopper and controller used are ThorLabs Model MC1000. The wheel was configured for 2 cycles (i.e. 2 open regions and 2 blocked regions); hence a 1 Hz drive produces a 2 Hz response. The Figure below shows the results of the time response testing for all 5 of the detectors in the ACME Radiometer Assembly.

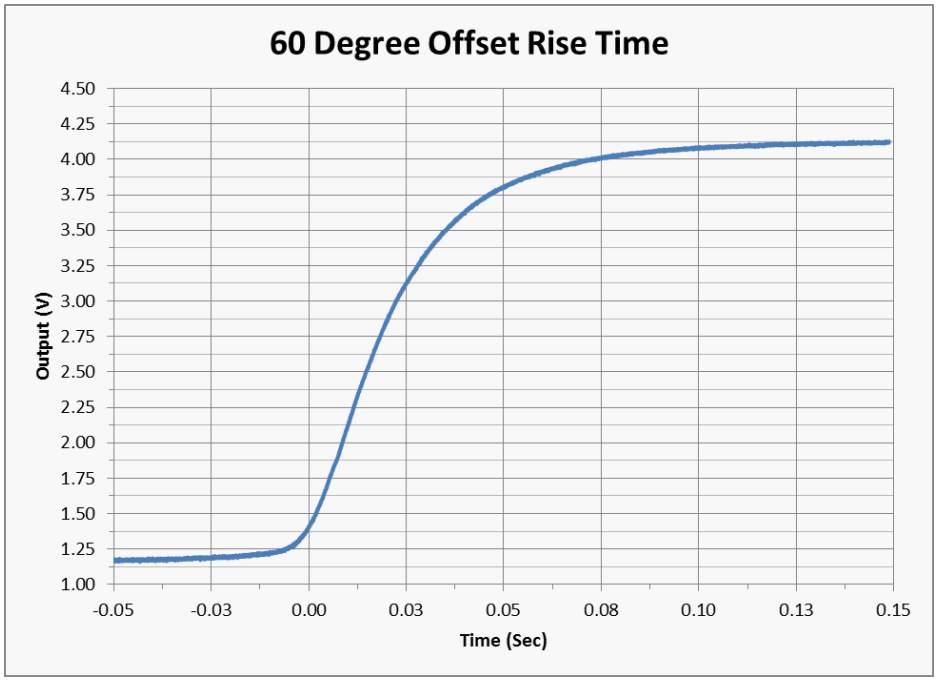


FIGURE 20. 60 Degree Offset Time Response

The table below shows the calculated response time for the detector based on the method above.

TABLE XXII. 60 Degree Offset Radiometer Response Time

Offset	V_{Out}	Time (s)
Min V	1.16	-0.04324
Max V	4.13	
Delta V	2.97	
62.50%	2.27	0.01180
Time Delta		0.05500
Response Time		55.02 ms

6.4 Statistics & Error Analysis

The accuracy is determined by a combination of random and systematic errors accounted for in the accompanying statistical analysis. This analysis represents the graphical data by method of least squares. Note the calculated Irradiances in the plots are all subject to the uncertainties and the Output Voltages recorded in the calibration also have measurement uncertainties.

The graphical approach used does not inherently account for unknown systematic errors.

The Table below summarizes the results of the graphical analysis of the collected data for the detector in the 60 Degree Offset Radiometer Assembly.

TABLE XXIII. 60 Degree Offset Statistical Analysis

Radiometer Statistics		60 Offset
Voltage Range	(V)	3.38
Min Voltage	(V)	1.27
Max Voltage	(V)	4.65
Irradiance Range	(W/cm^2)	0.0060356
Min Irradiance	(W/cm^2)	0.0000185
Max Irradiance	(W/cm^2)	0.0060541
Uncertainty	(± V)	4.8038E-06
(Linear) R^2		**0.9996**
(Linear) Slope		0.001798
(Linear) Intercept		-0.002307

The Table below shows the final Total Error at the Minimum Irradiance and Maximum Irradiance values, as well as at values representing 5%, 10% and 50% of the Max value.

In addition, the Irradiance at the 5% Error threshold has been calculated and is displayed.

TABLE XXIV. 60 Degree Offset Accuracy & Error Analysis

Accuracy/Error Summary	60 Offset
Error at Min	7.51%
Error at 5% Max	1.50%
Error at 10% Max	0.75%
Error at ½ Max	0.15%
Error at Max	0.08%
Min Irradiance @ 5% Error	0.0000961

APPENDIX A ACRONYMS AND ABBREVIATIONS

ACME	Advanced Combustion via Microgravity Experiments
BB	Broadband
CIR	Combustion Integrated Rack
COTS	Commercial off The Shelf
DC	Direct Current
ERD	Engineering Requirements Document
ESD	Electrostatic Discharge
FCF	Fluids and Combustion Facility
FWHM	Full Width at Half Max
FOV	Field of View
FS	Fused Silica
GL	Gray Level
IR	Infrared
ISS	International Space Station
ms	Milliseconds
N/A	Not Applicable
NASA	National Aeronautics and Space Administration
NB	Narrowband
ns	Nanoseconds
QE	Quantum Efficiency
SpaceDOC	Space Flight Systems Development and Operations Contract
UV	Ultraviolet
VDC	Volts (direct current)